

The Calculation of A^{-1} : The Gauss–Jordan Method

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Inverse of a Matrix

For a square matrix A , the inverse A^{-1} satisfies

$$AA^{-1} = I$$

where I is the identity matrix.

The Gauss–Jordan Method is a systematic way to compute A^{-1} .

If we take it column by column, we get

$$\begin{aligned} Ax_1 &= e_1 \\ Ax_2 &= e_2 \\ Ax_3 &= e_3 \end{aligned}$$

$$\text{where } e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Example Matrix

Example of a 3×3 matrix

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}$$

We want to find A^{-1} .

The equation becomes

$$A \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Systems of Equations

This gives three systems of equations

$$Ax_1 = e_1$$

$$Ax_2 = e_2$$

$$Ax_3 = e_3$$

- All systems have the same coefficient matrix A .
- Only the right-hand sides differ.

Gauss–Jordan Method

Instead of solving each system separately:

- Apply row operations to all systems simultaneously.
- Continue elimination until the left side becomes the identity matrix.

Steps:

- ✓ Create the augmented matrix $[A|I]$
- ✓ Perform row operations
- ✓ Transform A into I

Final Result

When the process is complete:

$$[A|I] \rightarrow [I|A^{-1}]$$

- Left side becomes Identity Matrix
- Right side becomes Inverse Matrix A^{-1}

Key Points:

- ✓ Gauss–Jordan method finds the inverse of a matrix.
- ✓ Works only for square matrices.
- ✓ If the matrix cannot be reduced to the identity matrix, the inverse does not exist.

Finding the Inverse of a Matrix using the Gauss–Jordan Method

Example: 1

Find the inverse of matrix A by using the Gauss-Jordan Method starting with $[A|I]$:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

Given matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

To find A^{-1} we start with the augmented matrix $[A|I]$.

Step 1: Form Augmented Matrix

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 2 & 0 & 1 & 0 \\ 1 & 2 & 3 & 0 & 0 & 1 \end{array} \right]$$

- Step 2: Convert the augmented matrix into Reduced echelon form by using elementary row operations.

Make zeros below first Pivot

$$\begin{aligned} R_2 &= R_2 - R_1 = (1, 2, 2 \mid 0, 1, 0) - (1, 1, 1 \mid 1, 0, 0) \\ &= (0, 1, 1 \mid -1, 0, 0) \end{aligned}$$

$$\begin{aligned} R_3 &= R_3 - R_1 = (1, 2, 3 \mid 0, 0, 1) - (1, 1, 1 \mid 1, 0, 0) \\ &= (0, 1, 2 \mid -1, 0, 1) \end{aligned}$$

Result:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{array} \right]$$

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Eliminate below second Pivot

$$R_3 = R_3 - R_2$$

Result:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array} \right]$$

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Eliminate above third Pivot

$$R_2 = R_2 - R_3$$

$$R_1 = R_1 - R_3$$

Result:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 1 & -1 \\ 0 & 1 & 0 & -1 & 2 & -1 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array} \right]$$

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Eliminate above second Pivot

$$R_1 = R_1 - R_2$$

Result:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -1 & 0 \\ 0 & 1 & 0 & -1 & 2 & -1 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array} \right]$$

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Left side becomes the identity matrix.

Therefore the inverse matrix is:

$$A^{-1} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

Verify $AA^{-1} = I$

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Example: 2

Using the Guass-Jordan Method, show that the matrix

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 1 & -2 & 1 \\ 5 & -2 & -3 \end{bmatrix}$$

has no inverse.

Condition for Matrix Inverse

A square matrix A has an inverse if

$$[A|I] \rightarrow [I|A^{-1}]$$

after performing Gauss–Jordan row operations.

If the left side cannot be reduced to the identity matrix, then

A^{-1} does not exist.

Such a matrix is called a singular matrix.

Form the Augmented Matrix

Start with

$$\begin{array}{c} [A|I] \\ \left[\begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 & 1 & 0 \\ 5 & -2 & -3 & 0 & 0 & 1 \end{array} \right] \end{array}$$

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Eliminate First Column

Row operations:

$$\begin{aligned}R_2 &= R_2 - R_1 \\R_3 &= R_3 - 5R_1\end{aligned}$$

Result:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 0 & -4 & 4 & -1 & 1 & 0 \\ 0 & -12 & 12 & -5 & 0 & 1 \end{array} \right]$$

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Next operation:

$$R_3 = R_3 - 3R_2$$

Result:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -3 & 1 & 0 & 0 \\ 0 & -4 & 4 & -1 & 1 & 0 \\ 0 & 0 & 0 & -2 & -3 & 1 \end{array} \right]$$

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The third becomes

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

This means the left side cannot become the identity matrix, because the identity matrix must be

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

But we obtained a row of zeros.

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Since the matrix cannot be reduced to the identity matrix A^{-1} does not exist.

Therefore,

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 1 & -2 & 1 \\ 5 & -2 & -3 \end{bmatrix}$$

is a singular matrix.